

COURSEMATERIAL

SUBJECT	ENGINEERINGPHYSICS(PH23ABS101)
UNIT	V
COURSE	B.TECH
DEPARTMENT	S&H
SEMESTER	II
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1.CourseObjectives

The objective of this course are to:

Bridge the gap between the Physics in school at 10+2 level and UG level engineering courses by identifying the importance of the optical phenomenon like interference, diffraction etc, enlightening the periodic arrangement of atoms in crystalline solids and concepts of quantum mechanics, introduce novel concepts of dielectric and magnetic materials, physics of semiconductors.

1. Prerequisites

Students should have knowledge on

1. WaveOptics
2. Crystallography
3. QuantumMechanics
4. Semiconductors

2. Syllabus

Semiconductors: Formation of energy bands – classification of solids -
 Intrinsic semiconductors: Density of charge carriers – Electrical conductivity –
 Fermi level – Extrinsic semiconductors: density of charge carriers (Qualitative)
 – dependence of Fermi energy on carrier concentration and temperature - Drift
 and diffusion currents – Einstein's equation – Hall effect and its applications -
 Direct and indirect bandgap semiconductors and its applications.

3. Courseoutcomes

1. CO 1: Understand the intensity variation of light due to interference, diffraction and polarization.
2. CO 2: Apply the basic concepts of crystal structures and X-ray diffraction to study the behavior of materials for engineering applications.
3. CO 3: Summarize the fundamental properties of dielectric and magnetic materials for engineering applications.
4. CO 4: Analyze the properties of quantum particles to interpret the energy band formation and classification of solids
5. CO 5: Assess the current flow mechanism to understand the transport phenomenon of charge carriers in semiconductors.

4. Co-PO/PSOMapping

APPLIED PHYSICS	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	PO11	PO12	PSO 1	PSO 2
CO1	3	3												
CO2	3	3												
CO3	3	3												
CO4	3	3												
CO5	3	3												

5. LessonPlan

Lecture No.	Weeks	Topicstobecovered	References
1	1	IntroductiontoSemiconductors	T1, R1
		Formation of energy bands	T1, R1
2		Intrinsicsemiconductors	T1, R1
3		Extrinsicsemiconductors(n-typeandp-type)	T1, R1

4		Derivation of hole density (p) and position of Fermi energy level	T1, R1
5	2	Electrical conductivity	T1, R1
6		Direct and Indirect band gap semiconductors	T1, R1
7		Hall effect - Hall coefficient - Applications of Hall effect	T1, R1
8		Drift and Diffusion currents	T1, R1
9	4	Problems-2	T1, R1
10		Problems-1	T1, R1

2. Activity Based Learning

1. Determination of the energy gap of a semiconductor

3. Lecture Notes

8.1. Semiconductors:

Formation of energy bands in solids

- We know that in a real solid, there are a very large number of atoms closely packed together.
- An isolated atom consists of positively charged nucleus and negatively charged electrons revolve around the nucleus in various orbits as shown in fig(1).
- In the innermost orbits, some electrons are tightly bonded with nucleus are called as bound electrons or inner electrons.
- In the outermost orbits, some electrons are loosely bounded with nucleus are called as valence electrons and the attraction force between the nucleus and these valence electrons will be very less. So that the electrons can be easily detached from their outermost orbits and become free and hence these electrons are known as free electrons.
- Let us consider the case of sodium atom with atomic number 11 and mass number 23. There are 11 electrons. Two electrons revolve in the first orbit, eight electrons in the second orbit and one electron in the third orbit. The first, second, third orbits are also known as K, L, M orbits respectively.

Each orbit has a amount of

energy associated with it. i.e., each electron in an orbit will possess discrete energy is called energy level.

- A convenient way of representing the different energy orbits of electron in an isolated atom is known as energy level diagram as shown in fig(2).
- The first orbit represents the first energy level; the second orbit represents the second energy level and so on.
- When large numbers of atoms are closely packed in a solid, then their corresponding energy levels are closely packed together and form a region called energy bands. Hence, energy band can be defined as the range of energies possessed by an electron in a solid or the large number of discrete but closely spaced energy levels.
- A convenient way of representing the different energy bands of electron in a solid is known as energy band diagram as shown in fig(2).
- The range of energies possessed by bound electrons or inner electrons in a solid is called inner bands.
- The range of energies possessed by valence electrons or outermost electrons in a solid is called valence band. This band is completely or partially filled.
- The range of energies possessed by free electrons or conduction electrons in a solid is called conduction band.

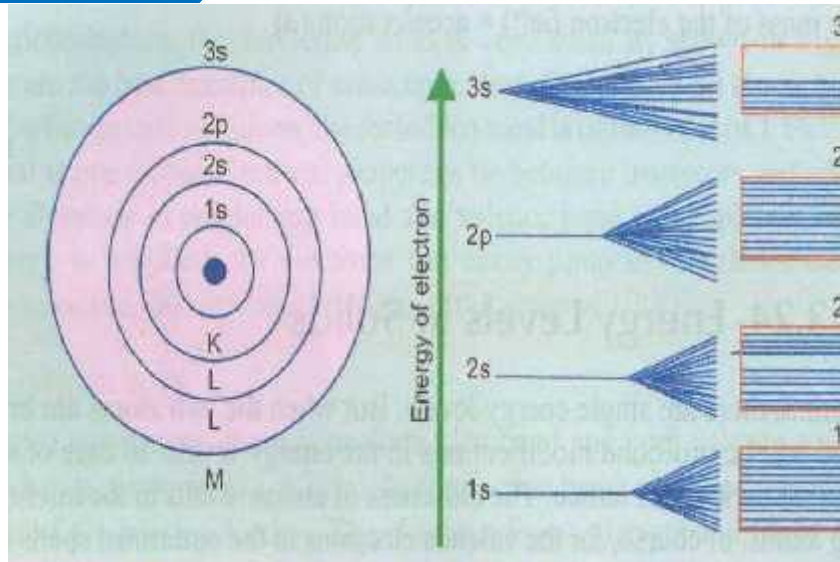
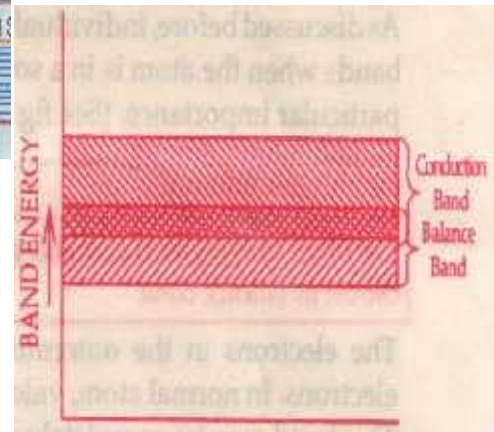


Fig:Energy levels and bands in sodium material.



- Energy bands are separated by regions called forbidden energy bands.
- The separation between conduction band and valence band on the energy level diagram is known as forbidden energy gap.
- No electron can allow or stay in this region, because there are no energy levels in this region.
- Conductors are those substances which easily allow the passage of electric current through them. It is because of large number of free electrons available in a conductor.
- Examples: Metals (silver, copper and aluminum etc).
- In terms of band theory, the valence band and conduction band overlap each other as shown in fig.
- So no energy is required to move the electron from valence band to conduction band.

Differences between Conductors, semiconductors and Insulators

Conductors

- Free electrons are available in conduction band and thus current flows through it, when a potential difference is applied across it.
- When the temperature increases, then the electrical resistance increases due to collision of free electrons with each other.
- So, conductors have positive temperature coefficient of resistance.

Insulators:

- Insulators are those substances which do not allow the flow of electric current through them.
- Examples are dry wood, glass and diamond etc.
- In terms of band theory, the valence band is full while the conduction band is empty. Further the energy gap between valence band and conduction band is very large, i.e., greater than 5 eV.
- So electrons cannot jump from the valence band into the conduction band.
- Therefore, a very high electric field is required to jump electrons from the valence band into the conduction band.
- For this reason the electrical conductivity is extremely small and may be nil under ordinary conditions.
- At room temperature, the electrons in the valence band don't have sufficient energy to cross



crossover to the conduction band. However, when the temperature increased, some of the electrons in the valence band may acquire sufficient energy to crossover to the conduction band.

Hence, the resistance of an insulator decreases with increase in temperature i.e.,

an

insulator has negative temperature coefficient of resistance.

Semiconductors:

- Semiconductors are those substances whose electrical conductivity lies in between conductors and insulators.
- Examples: Si, Ge, GaAs, ZnS and GaP etc.

- In terms of band theory, the valence band is almost filled and conduction band is almost empty. Further the energy gap between valence band and conduction band is very small (0.78 eV for Ge and 1.21 eV for Si etc)
- So electrons can easily jump from the valence band into the conduction band.
- Therefore, a very small electric field is required to jump electrons from the valence band into the conduction band.
- At 0 K, the valence band is completely full and conduction band is completely empty. Therefore, semiconductor virtually behaves as an insulator at low temperatures. However, even at room temperature, some electrons (about one electron for 10^{10} atoms) in the valence band may acquire sufficient energy to cross over to the conduction band imparting little conductivity to the semiconductor.

As the temperature increased, more electrons in the valence band cross over to the conduction band and conductivity increases. Hence, the resistance of semiconductors decreases with increase in temperature i.e., semiconductors have a negative temperature coefficient of resistance.

Semiconductor definition

A Semiconductor is a substance whose conductivity or resistivity (10^{-4} to 0.5 ohm-m) lies in between conductors and insulators.

Examples: Si, Ge, GaAs, AlAs and InP etc.

Properties:

- They have crystalline structure.
- They are formed by covalent bonds.
- They have a negative temperature coefficient of resistance.
- At $T = 0 \text{ K}$, they behave as insulators.
- When the impurities (III group or V group elements) added to the pure semiconductor, then

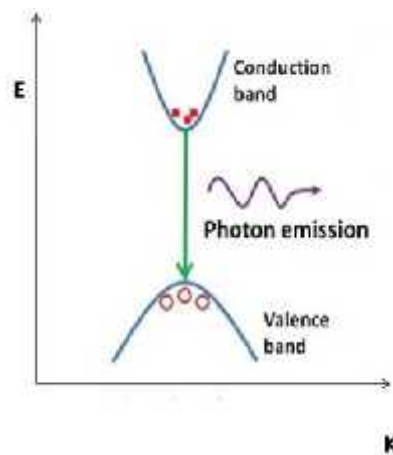
electrical conductivity increases.

8.2. Direct Band Gap And Indirect Band Semiconductors: Based on the type of energy emission, these semiconductors can be of two groups.

1. Direct band gap semiconductors
2. Indirect band gap semiconductors

Direct Band Gap Semiconductors:

- The semiconductor, in which the minimum of the conduction band lies directly above the maximum of the valence band in the E-K space diagram (as shown in fig), is known as a direct band gap semiconductor.



- In a direct band gap semiconductor such as GaAs, AlAs and InP, when an excited electron falls back into the valence band, the electrons and holes recombine to produce light energy.

i.e., $e^- + \text{hole} \longrightarrow h\nu (\text{photon})$

- This process is known as radiative recombination and also called spontaneous emission.
- These direct band gap semiconductors are used to make LED's and lasers of different colours.

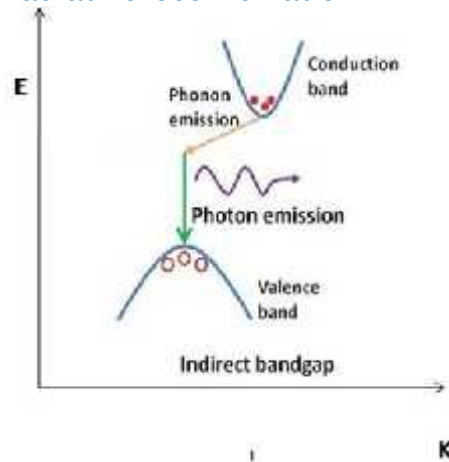
Indirect Band Gap Semiconductors:

- This semiconductor, in which the minimum of the conduction band does not lie directly above the maximum of the valence band in the E-K space diagram (as shown in fig), is known as an indirect band gap semiconductor.

- In an indirect gap semiconductor such as Si, Ge and Gap, when an excited electron falls back into the valance band, the electrons and holes recombine to generate heat and is dissipated within the material.

i.e., $\bar{e} + \text{hole} \longrightarrow \text{phonon}$

- This process is known as non-radiative recombination.



8.3. Intrinsic semiconductors and extrinsic

semiconductors Based on purity semiconductors are classified into types,

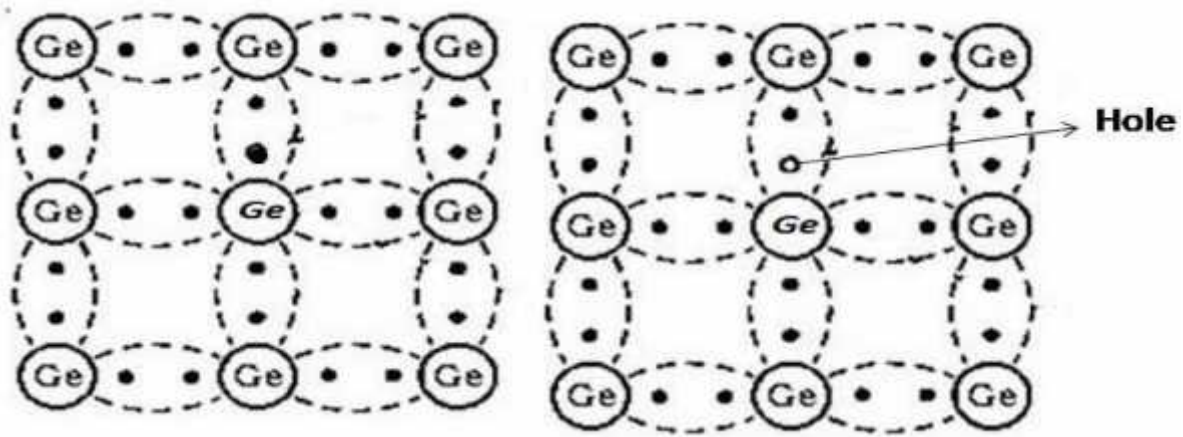
1. Intrinsic semiconductors
2. Extrinsic semiconductors

8.3.1. Intrinsic semiconductor:

A semiconductor in an extremely pure form is known as an intrinsic semiconductor. Examples: Si, Ge etc.

Explanation:

- Consider a pure (semiconductor) Si or Ge Crystal. We know that Ge is a fourth group element, i.e., it has four valence electrons; to get stability each silicon atom will make four covalent bonds with neighboring silicon atoms as shown in fig.
- At absolute zero temperature (0K): all the electrons are tightly bound by these semiconductor atoms and the covalent bonds are very strong. So, there are no free electrons. Therefore, the semiconductor crystal behaves as a perfect insulator at absolute zero temperature.

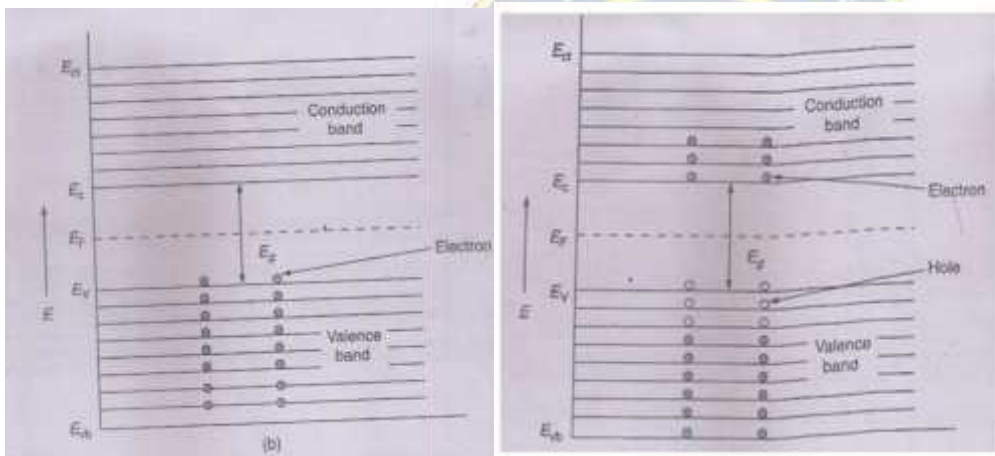


a) At $T=0K$

b) At $T > 0K$

- Above absolute zero ($>0K$): When the temperature is raised, some of the covalent bonds in the semiconductor break due to the thermal energy supplied.
- The breaking of bonds set those electrons free, at the same time vacant sites produced in the covalent bond called "hole".
- The result is that a few free electrons exist in the semiconductor. These free electrons can constitute a tiny electric current if potential difference is applied across the semiconductor as shown in fig(c)

- Energybanddescription: Fig (d) shows the energy band diagram. As the temperature is raised, some of the valence electrons acquire sufficient energy to enter into the conduction band and thus become free electrons.
- Under the influence of electric field, these free electrons will constitute electric current. It may be noted that each time a valence electron enters into the conduction band; a hole is created in the valence band, which acts as a positive charge.
- For one electron set free, one hole is created. Therefore, thermal energy creates hole-electron pairs;
- In an intrinsic semiconductor, even at room temperature, electron-hole pairs are recreated. Every free electron creates one hole in the valence band.
- Therefore, the number of electrons in the conduction band and the number of holes in the valence band become equal.

(a) At $T=0K$ (b) At $T>0K$

- When electric field is applied across an intrinsic semiconductor, the current conduction takes place by two processes namely: by free electrons and holes.
- Therefore, the total current inside the semiconductor is the sum of currents due to free electrons and holes

$$I = I_{\text{Electron}} + I_{\text{Hole}}$$

8.3.2. Extrinsic Semiconductor:

The electrical conductivity of a pure semiconductor is very small. To increase the conductivity, impurities are added. The impurity added semiconductor is called "Extrinsic Semiconductor". (OR)

The conductivity of intrinsic semiconductor can be increased enormously by adding small amounts of impurity atoms [such as III or V group atoms]. This impure semiconductor is called "Extrinsic Semiconductor".

Doping:

The process of adding impurities to a semiconductor is called doping and the impurity added is called "Dopant".

Depending upon the type of impurity added, Extrinsic Semiconductors are classified into two types.

1. N-Type Semiconductor.
2. P-Type Semiconductor

8.3.3. N-Type Semiconductor:

When a small quantity of pentavalent impurity is added to a pure semiconductor, it is known as 'n-type semiconductor'.

Explanation:

- To explain the formation of n-type semiconductor, consider a pure (semiconductor) Germanium Crystal.
- We know that germanium atom has four valence electrons, when a small number of pentavalent impurities like arsenic (As) is added to a germanium crystal, a large number of free electrons are available in the crystal.
- The reason is simple. Arsenic is pentavalent i.e., its atom has five valence electrons. The four of the five valence electrons of Arsenic will make covalent bonds with four germanium atoms. The fifth valence electron of arsenic atom finds no place in covalent bonds and is thus free as shown in fig.
- Therefore, for each arsenic atom added, one free electron will be available in the germanium crystal.

- Thus, a small number of arsenic impurities (As) are added to a germanium crystal, a large number of free electrons are available in the crystal.
- Pentavalent impurities are also known as donor impurities because they donate free electrons to semiconductor crystal.

Energy Band Description:

- Fig(2) shows the energy band description of n-type semiconductor.
- The addition of donor impurity to an intrinsic semiconductor creates extra energy level (called donor energy level) just below the bottom of conduction band.
- At Absolute Zero: At absolute zero temperature the fifth electrons of arsenic atoms occupy the donor level (represented by dashed lines) as shown in fig.
- At above Absolute Zero: Even room temperature provides enough thermal energy to push the electrons sitting at donor energy level to conduction band. (The result is that at room temperature, there are a large number of free electrons in the conduction band as shown in fig) and also few of the covalent bonds in the semiconductor break due to the thermal energy applied.
- Due to the breaking of covalent bond, there will be a few holes formed in the valence band at this temperature. The result is that at room temperature, there are a large number of free electrons in the conduction band compared to the holes in valence band, i.e., Electrons are majority carriers and holes are minority carriers. So this material is called n-type semiconductor.

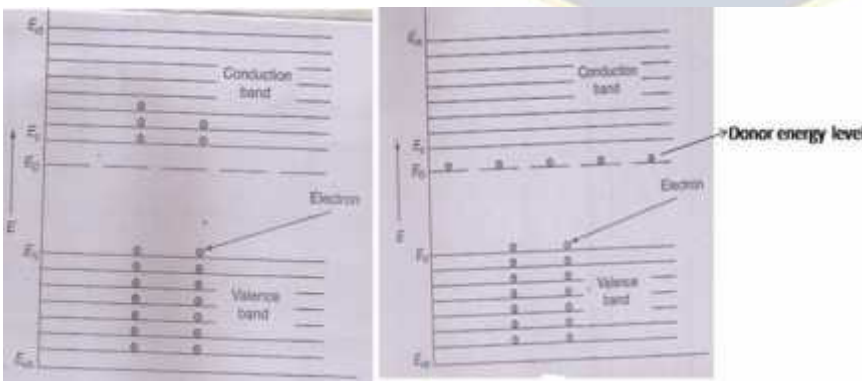


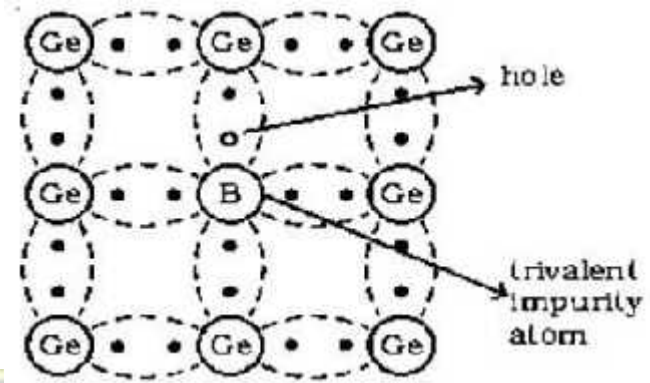
Fig: a) At $T=0K$ b) At $T>0K$

8.3.4. P-Type Semiconductor:

When a small quantity of trivalent impurity (such as indium (In), Boron (B), aluminum (Al) etc.) is added to a pure semiconductor, it is called P-type semiconductor.

Explanation:

- To explain the formation of p-type semiconductor, consider a pure (semiconductor) Germanium Crystal.
- We know that germanium atom has four valence electrons, when a small amount of trivalent impurities like boron atoms (B) are added to a germanium crystal, a large number of holes are available in the crystal.
- The reason is simple. Boron is trivalent i.e.; its atom has three valence electrons. The three valence electrons of Boron atom will make three covalent bonds with surrounding Ge atoms.
- The fourth bond is incomplete because it is short of one electron. This missing electron is called a "hole".
- Therefore, for each boron atom added, one hole will be available in the germanium crystal.
- Thus, a small amount of boron impurities (B) are added to a germanium crystal, a large number of free holes are available in the crystal.
- Trivalent dopant impurities are also called acceptors. Because they have need of extra electrons to complete their fourth covalent bond formation with Ge atoms.
- Energy Band Description: Fig (2) shows the energy band description of p-type semiconductor. The addition of acceptor impurity to an intrinsic semiconductor creates an extra energy level (called acceptor energy level) just above the top of the valence band [see fig].
- Since the energy difference between acceptor level and valence band energy level is very small so even at low temperature the valence band electrons gain sufficient thermal energy to occupy the acceptor energy level states as shown in fig.
- After receiving an electron, the Boron atom becomes a negative ion.
- But at room temperature provides enough thermal energy to push the electrons in the valence band to the acceptor energy level and also due to thermal agitation a few electrons jump from V.B to C.B.



- The result is that at room temperature there are a large number of holes in the V.B compared to the electrons in the C.B. i.e., holes are majority carriers and electrons are minority carriers. So, this material is called P-type semiconductors.

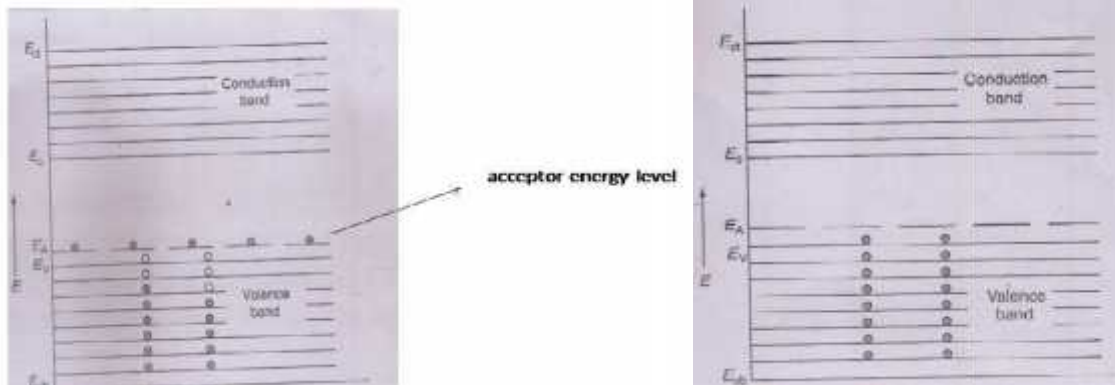


Fig: a) At $T=0K$

b) At $T>0K$

8.4. Electrical conduction in semiconductors

In these semiconductors electrical conduction arises due to two processes namely

1. Drift and
2. Diffusion

8.4.1. Drift:

Definition:-

Under an applied electric field both the charge carriers i.e., electrons and holes acquire an average velocity called the drift velocity (V_d) and this process is called drift.

The drift velocity V_d is small. It is directly proportional to the electric field E . i.e., $V_d \propto E$

$$V_d = \mu E \longrightarrow (1)$$

Where μ is a constant and is called the mobility of charge carriers.

If μ_n and μ_p are the electron mobility and hole mobility respectively, then $V_n = \mu_n E$

and $\longrightarrow (2)$

$$V_p = \mu_p E \longrightarrow (3)$$

1. We know that current $I = neAV_d$. But in a semiconductor, the total current is the sum of electron current and hole current.

Let n be the electron concentration and p be the hole concentration.

Therefore $I = I_n + I_p$

$$= neAV_n + pAV_p$$

$$= eA(nV_n + pV_p)$$

[From equations (2) & (3)]

$$= eA(n\mu_n E + p\mu_p E)$$

$$= eA(n\mu_n + p\mu_p)E$$

2. Current density $J = \frac{I}{A}$

$$J_n = \frac{neA\mu_n E}{A} \quad \text{and}$$

$$J_p = \frac{peA\mu_p E}{A}$$

Therefore, total current density $J = J_n + J_p$

$$= \frac{eA(n\mu_n + p\mu_p)E}{A} = e(n\mu_n + p\mu_p)E$$

$$\text{Conductivity } \sigma = \frac{J}{E} = \frac{e(n\mu_n + p\mu_p)E}{E} = e(n\mu_n + p\mu_p)$$

NOTE:-

1. For an intrinsic semiconductor: $n = p = n_i$

Where n_i is the carrier concentration of the intrinsic semiconductor.

$$I_{inc} = eA(n_i\mu_n + n_i\mu_p)E$$

$$= eAn_i(\mu_n + \mu_p)E$$

$$\square \text{ Current density } J_{in} = \frac{I_{in}}{A} = \frac{eAn_i(\mu_n + \mu_p)E}{A} = en_i(\mu_n + \mu_p)E$$

$$\text{Conductivity } \sigma_{in} = \frac{J_{in}}{E} = \frac{en_i(\mu_n + \mu_p)E}{E} = en_i(\mu_n + \mu_p)$$

FOR AN EXTRINSIC SEMICONDUCTOR:-

a) For n-type semiconductor:-

The hole current is neglected, as the holes are minority carriers and the electrons are the majority carriers.

$n \approx N_d$ is the donor electron concentration

[Where N_a can be neglected] $I =$

$$eAV_n N_d$$

$$= eA\mu_n E N_d$$

$$J = \frac{I}{A} = e\mu_n E N_d$$

$$\sigma = \frac{J}{E} = e\mu_n N_d$$

b) For p-type semiconductor:-

The electron current is neglected and $p = N_a$, where N_a is the acceptor hole concentration.

[When N_d can be neglected].

$$I = eAV_p N_a$$

$$= eA\mu_p E$$

$$N_a J = \frac{I}{A} = e\mu_p E N_a$$

$$\sigma = \frac{J}{E} = e\mu_p N_a$$



8.4.2. DIFFUSION:-

Diffusion takes place due to existence of non-uniform concentration of charge carriers.

Definition:-

Due to non-uniform carrier concentration in a semiconductor, the charge carrier's moves from a region of higher concentration to a region of lower concentration.

This process is known as diffusion of charge carriers.

→ According to Fick's law, the diffusion flux F is directly proportional to concentration

gradient <i.e. carrier concentration per unit length > of the particles and is in the opposite direction;

$$F_a = -\frac{\partial(\Delta N)}{\partial x}$$

$$F = -D \frac{\partial(\Delta N)}{\partial x} \quad (1) \rightarrow$$

Where D is diffusion constant.

And ΔN is the excess carrier concentration.

Let Δn and Δp be the excess concentration in a semiconductor material. Then Fick's

$$F = -e D_n \frac{\partial(\Delta N)}{\partial x} \quad (\text{For electrons})$$

$$F = -D_n \frac{\partial(\Delta N)}{\partial x} \quad (2) \rightarrow$$

and $F = -e D_p \frac{\partial(\Delta P)}{\partial x} \quad (\text{For electrons})$

$$F = -D_p \frac{\partial(\Delta P)}{\partial x} \quad (3) \rightarrow$$

→ But the current density due to diffusion depends on charge of particle and rate of diffusion.

□ Current density (J) = Charge of particle × Rate of diffusion → (4)

→ The diffusion flux density due to electron is given by

$$J_n(\text{Diff}) = -e \left[-D_n \frac{\partial(\Delta N)}{\partial x} \right] \quad \text{from (4)}$$

$$J_n(\text{Diff}) = e D_n \frac{\partial(\Delta N)}{\partial x} \quad (5)$$

The diffusion flux density due to hole is given by

$$J_p(\text{Diff}) = +e \left[D_p \frac{\partial(\Delta P)}{\partial x} \right] \quad (6) \rightarrow$$

Where $+e$ and $-e$ are charge of the hole and electron respectively.

□ The total current density due to diffusion is given by

$$J_{\text{TOTAL, (DIFF)}} = J_n(\text{diff}) + J_p(\text{diff})$$

$$J_{\text{TOTAL (DIFF)}} = e D_n \frac{\partial(\Delta N)}{\partial x} + e D_p \frac{\partial(\Delta P)}{\partial x} \quad (7) \rightarrow$$

NOTE:-

The total current density due to excess electrons is the sum of the current densities due to drift and diffusion of electrons.

$$J_n = J_n(\text{drift}) + J_n(\text{diff})$$

$$= \Delta n \mu_n E + e D_n \frac{\partial(\Delta N)}{\partial x} \quad (8) \rightarrow$$

Similarly,

The total current density due to excess holes is the sum of the hole densities due to drift and diffusion of holes.

$$J_p = \Delta p e \mu_p E - e D_p \frac{\partial(\Delta p)}{\partial x} \quad (9)$$

8.5. EINSTEIN'S RELATION:-

Einstein showed the direct relation between the mobility (μ) and diffusion coefficient (D) of a semiconductor. This relation is called Einstein relation.

At any disturbance in equilibrium, the drift and diffusion current densities due to carrier concentration are equal.

$$\text{i.e., } J_{n(\text{drift})} = J_{n(\text{diff})}$$

$$(\Delta n) e E \mu_n = D_n e \frac{\partial(\Delta n)}{\partial x} \quad (1)$$

The force on excess electrons (Δn) to restore equilibrium is given by the product of Δn and electric force eE on each electron. Thus,

$$F = (\Delta n) e E \quad (2)$$

Equation (1) can be written

$$\text{as } F \cdot \mu_n = D_n e \frac{\partial(\Delta n)}{\partial x}$$

$$F = \frac{D_n}{\mu_n} e \frac{\partial(\Delta n)}{\partial x} \quad (3)$$

From the kinetic theory of gases, the force on gas molecules (charge carriers) is given by

$$F = K_B T \frac{\partial(\Delta n)}{\partial x} \quad (4)$$

Comparing (3) and (4), we get

$$K_B T \frac{\partial(\Delta n)}{\partial x} = \frac{D_n}{\mu_n} e \frac{\partial(\Delta n)}{\partial x}$$

$$K_B T = \frac{D_n}{\mu_n} e$$

$$D_n = \frac{\mu_n K_B T}{e}$$

$$\frac{D_n}{\mu_n} = \frac{k_B T}{e} \quad (5) \quad \rightarrow$$

Similarly, for holes,

$$\frac{D_p}{\mu_p} = \frac{k_B T}{e} \quad (6) \quad \rightarrow$$

From equations (5) and (6)

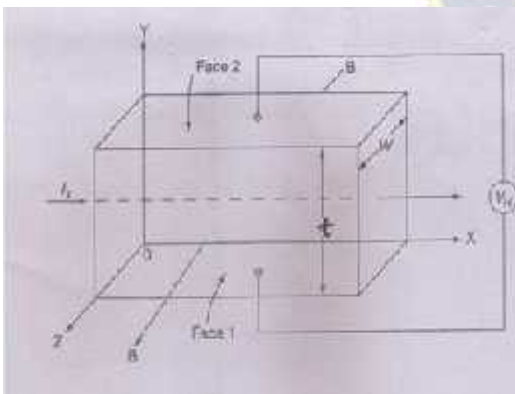
$$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p}$$

$$\frac{D_n}{D_p} = \frac{\mu_n}{\mu_p}$$

The above relation is known as Einstein's relation.

8.6. Hall Effect:

When a current carrying semiconductor is placed in the magnetic field, then a potential difference (voltage) or electric field is developed in the material perpendicular to both the current and the applied magnetic field. This effect is known as Hall Effect and the developed potential difference (voltage) is known as Hall voltage.



Explanation:

- Consider a semiconductor slab of thickness 't' and width 'w' in which current (I) flowing along X-direction and magnetic field is applied along Z-direction as shown in fig.
- The charge carriers inside the semiconductor experience a force due to the applied magnetic field called Lorentz force.

- As a result, electrons move (drift) towards downward direction with velocity ' V_d ' and accumulate at bottom surface (face 1) and the corresponding positive charge on upper surface (face 2) of the material as shown in fig.
- Thus a potential difference and electric field developed across the surface. This developed electric field along Y-direction exerts a force ($F_e = eE_H$) on the electrons to prevent further accumulation at the bottom surface.
- Therefore, the force due to potential difference $F_e =$

eE_H The force due to magnetic field $F_L = BeV_d$

At steady

state $F_e = F_L$

$$eE_H = BeV_d$$

$$E_H = BV_d \quad (1)$$

Forn-type:

If ' n ' is the electrons carrier concentration in the semiconductor, then current density $J = -neV_d$

$$V_d = \frac{J}{-ne} \quad (2)$$

Substituting eq (2) in eq (1)

$$E_H = \frac{BJ}{-ne}$$

(3)

$$E_H = R_H BJ$$

Where $R_H = \frac{1}{-ne}$ is known as Hall coefficient.

For p-type:

If ' p ' is the electrons carrier concentration in the semiconductor, then current density $J = peV_d$

$$V_d = \frac{J}{pe} \quad (2)$$

Substituting eq (2) in eq (1)

$$E_H = \frac{BJ}{pe}$$

$$E_H = R_H BJ \quad (3)$$

Where $R_H = \frac{1}{pe}$ is known as Hall coefficient

8.6.1. Hall coefficient in terms of hall voltage:

If the thickness of the sample is 't' and the developed voltage is V_H , then the Hall voltage

$$V_H = E_H \cdot t \quad \longrightarrow (4)$$

Substituting eq(3) in eq(4);

$$V_H = R_H B J t \quad \longrightarrow (5)$$

If 'w' is the

width of the semiconductor, then area of the

sample $A = w \cdot t$

But current density

$$J = \frac{I}{A} = \frac{I}{wt} \quad (6)$$

Substituting eq(6) in eq(5); we get

$$V_H = \frac{R_H B I}{w}$$

$$R_H = \frac{V_H w}{B I}$$

$$R_H = \frac{V_H}{B I}$$

This is the relation between Hall coefficient and hall voltage.

- The electrical conductivity in a semiconductor is given by

$$\sigma = ne\mu$$

$$\mu = \frac{\sigma}{ne}$$

$$\mu_H = \frac{R_H}{\sigma}$$

8.6.2. APPLICATIONS:

- Knowing R_H and σ , the mobility of charge carriers can be measured. i.e., $\mu = R_H \sigma$
- It is used to find the carrier concentration

$$n = \frac{1}{e R_H \sigma} \quad \text{and} \quad p = \frac{1}{e R_H \sigma}$$

- It is used to determine whether the material is p-type or n-type semiconductor. i.e., if R_H is negative then the material is n-type.
if R_H is positive then the material is p-type.

4. PracticeQuiz

1. Thenumberofvalenceelectrons in Si atom is_____
 - a)1 b)2 c)3 d)4
2. Electricconduction in a semiconductor occurs due to the motion of_____
 - a) freeelectronsonly b)holesonly
 - c)bothfreeelectronsandholes d)neitherelectrons norholes
3. As compared to an intrinsic semiconductor, the resistivity of an extrinsic semiconductor is_____
 - a) higher b)lower
 - c)resistivity can be compared d)none
4. The carrier concentration in an intrinsic semiconductor _____ with increase of temperature.
 - a) increases b)decreases c)remains same d)none
5. To obtain p-type semiconductor, the following impurities doped to Si (or) Ge _____
 - a) arsenic b)antimony c)indium d)phosphorus
6. The majority charge carriers in p-type semiconductor _____
 - a) electrons b)holes c)positive ions d)negative ions
7. In intrinsic semiconductors, the Fermi level lies _____
 - a) near the conduction band b)near the valence band
 - c)at midway of energy gap d)none of these
8. The Fermi level in n-type semiconductor lies _____
 - a) near the valence band b)near the conduction band
 - c)exactly at middle of the energy gap d)none of these

9. The energy gap of semiconductor is _____

- a) constant b) varies with temperature
c) varies with voltage d) varies with doping concentration

10. If μ_e is $0.19 \text{ m}^2/\text{Vs}$ then the diffusion coefficient of electron in silicon at 300K is _____

- a) $4.9 \times 10^{-3} \text{ m}^2/\text{sec}$ b) $9.4 \times 10^3 \text{ m}^2/\text{sec}$
c) $4.9 \times 10^3 \text{ m}^2/\text{sec}$ d) $9.4 \times 10^{-3} \text{ m}^2/\text{sec}$

11. If the charge carriers are electrons, the Hall coefficient is _____

- a) positive b) negative c) zero d) none of these

13. The Hall coefficient, $R_H =$

- a) $\frac{1}{ne}$ b) $\frac{n}{e}$ c) $\frac{e}{n}$ d) en

14. The conductivity of silicon increases by _____

_____ times by adding 10 parts of boron per million parts of silicon.

- a) 10 b) 100 c) 1000 d) 10000

15. For intrinsic semiconductor at 300K, $n_i = 2.4 \times 10^{19} \text{ m}^{-3}$, μ_e and μ_p are $0.39 \text{ m}^2 \text{V}^{-1} \text{S}^{-1}$ and $0.19 \text{ m}^2 \text{V}^{-1} \text{S}^{-1}$ respectively, then the value of conductivity is _____

- a) $2.2 \Omega^{-1} \text{m}^{-1}$ b) $1.2 \Omega^{-1} \text{m}^{-1}$ c) $2.1 \Omega^{-1} \text{m}^{-1}$

d) $2.2 \Omega \text{m}$

16. In case of intrinsic semiconductors _____

- a) $E_F > \frac{E_C + E_V}{2}$ b) $E_F < \frac{E_C + E_V}{2}$ c) $E_F = \frac{E_C + E_V}{2}$ d) none of these

17. The law of mass action is _____

- a) $\frac{n}{p} = n_i^2$ b) $\frac{p}{n} = n_i^2$ c) $np = n_i^2$ d) none

18. The diffusion current is proportional to _____ of charge carriers.

- a) concentration gradient b) drift velocity

- c) mobility d) none

19. $\frac{D_n \mu_n}{D_p \mu_p}$, this is known as _____
- a) equation of continuity b) Einstein's relation
c) Hall relation d) none of these
20. Pick out the direct band-gap semiconductor.
- a) Si b) Ge c) GaAs d) none of these
21. The conductivity of a superconductor is _____
- a) infinite b) finite c) zero d) none of these
22. An example of superconductor is
- (a) Cu (b) Si (c) Ge (d) Hg
23. Type-I superconductors are _____
- a) soft superconductors b) hard superconductors
c) ductile superconductors d) flexible superconductors
24. Superconductivity mainly occurs due to _____
- a) electron-electron interaction b) electron-phonon interaction
c) proton-proton interaction d) none of these
25. BCS theory introduces the concept of _____
- a) phonon pairs b) Cooper pairs c) electron-hole pairs d) hole pairs

10. Assignment Questions:

S.No	Question	BL	CO
1	Explain Fermi level? Derive the expressions for intrinsic carrier concentration and Fermi level for intrinsic semiconductor.	2 & 3	5
2	Distinguish drift and diffusion processes in a semiconductor with the help of relevant expressions.	2	5
3	Explain Hall Effect and describe Hall Effect in a semiconductor along with its applications.	2	5

4	Define critical magnetic field? Distinguish between type I and type II superconductors	2	5
5	Define critical temperature? Describe the effects in a superconductor	1	5
6	Describe BCS theory of superconductivity and write the applications of superconductors.	2	5

11. Question & Answers

S.No		BL	CO
1	What is carrier concentration? The number of electrons or holes per unit area is called as carrier concentration.	1	5
2	Define diffusion? Due to non-uniform carrier concentration in a semiconductor, the carrier's move from a region of higher concentration to a region of lower concentration. This process is known as diffusion of charge carriers.	1	5
3	What is Hall Effect? When a current-carrying semiconductor is placed in a magnetic field, then a potential difference (voltage) or electric field is developed in the material perpendicular to both the current and the applied magnetic field. This effect is known as Hall Effect and the developed potential difference (voltage) is known as Hall voltage.	1	5



12. Questions

S.No	Question	BL	CO
1	Explain about intrinsic and extrinsic semiconductors with fermi energy levels.	2	5
2	Describe direct and indirect band gap semiconductors. Write the applications of semiconductors	2	5

13. Supportive Online Certification Courses

1. Engineering Physics I (Experimental) by Prof. Alike Khare Prof. Pratima Agarwal Prof. S. Raviconducted by IIT Guwahati - 12 weeks
2. Engineering Physics I (Theory) by Prof. Alike Khare Prof. Pratima Agarwal Prof. S. Raviconducted by IIT Guwahati - 12 weeks

14. Prescribed Text Books & Reference Books

Text Book

1. A text book of Engineering Physics –
Dr. M. N. Avadhanulu & Dr. P. G. Kshirsagar, S. Chand and Company, 11 Edition, 2019.
2. Engineering Physics – B. K. Pandey and S. Chaturvedi, Cengage Learning, 2013.

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1. Engineering Physics – Shatendra Sharma, Jyotsna Sharma, Pearson Education, 2018
2. Engineering Physics – K. Thyagarajan, McGraw Hill Publishers, 2019
3. Engineering Physics - Sanjay D. Jain, D. Sahasrambudhe and Girish, University Press, 2010.
4. Semiconductor physics and devices - Basic principle - Donald A. Neamen, McGraw Hill, 2011.
5. Solid State Physics, A. J. Dekker, Macmillan Education UK, 1969
6. Kittel's Introduction to Solid State Physics, Charles Kittel, Wiley India Edition Paperback, 2019